MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – TEST (SET – C)** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter: Areas Related to Circles** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory.
2. The question paper consists of 11 questions divided into three Sections: A, B, and C.
3. **Section A** comprises 5 Multiple Choice Questions (MCQs) of 1 mark each.
4. **Section B** comprises 3 Short Answer questions of 2 marks each.
5. **Section C** comprises 3 Long Answer questions of 3 marks each.
6. Use of calculator is not permitted.

SECTION A: Multiple Choice Questions (5 × 1 = 5 Marks)

Q1. The area of the sector of a circle of radius 12 cm is 60π cm^2. The central angle of this sector is: [1]

(A) $120°$ (B) $60°$ (C) $75°$ (D) $150°$

Q2. The area of a square inscribed in a circle of radius $2\sqrt{5}$ cm is: [1]

(A) 50 cm^2 (B) 100 cm^2 (C) 40 cm^2 (D) 80 cm^2

Q3. The perimeter of a sector of a circle of radius 21 cm which subtends an angle of $60°$ at the centre of the circle is: [1]

(A) 22 cm (B) 43 cm (C) 64 cm (D) 462 cm

Q4. The numerical value of the area of a circle is equal to that of the perimeter of a semicircular disc, both having equal radius r. The radius is: [1]

(A) 1 unit (B) 2 units (C) $\left(1+\frac{2}{\pi}\right)$ units (D) $\left(2+\frac{2}{\pi}\right)$ units

Q5. If the perimeter of a square is equal to that of a circle, then the ratio of their areas is: [1]

(A) 7 : 22 (B) 11 : 14 (C) 22 : 7 (D) 14 : 11

SECTION B: Short Answer Questions (3 × 2 = 6 Marks)

Q6. The shape of the top of a table is that of a major sector of a circle with centre O and central angle $\angle AOB = 90°$. If $OA = OB = 42$ cm, find the perimeter of the top of the table. $\left[\text{Use } \pi = \frac{22}{7}\right]$ [2]

Q7. A brooch is made with silver wire in the form of a circle with diameter 35 mm. The wire is also used in making 5 diameters which divide the circle into 10 equal sectors. Find the total length of the silver wire used in making the brooch. $\left[\text{Use } \pi = \frac{22}{7}\right]$ [2]

Q8. A circular pizza of diameter 35 cm is cut into 8 equal sectors. Find the perimeter of each slice of pizza. $\left[\text{Use } \pi = \frac{22}{7}\right]$ [2]

SECTION C: Long Answer Questions (3 × 3 = 9 Marks)

Q9. Reeti prepares a Rakhi for her brother Ronit. The Rakhi consists of a rectangle of length 8 cm and breadth 6 cm inscribed in a circle. Find the area of the shaded region (area of the circle excluding the rectangle). [Use $\pi = 3.14$] **[3]**

Q10. A circle of radius 12 cm is divided into 11 identical sectors. Find the total area of 7 such sectors. $\left[\text{Use } \pi = \frac{22}{7}\right]$ **[3]**

Q11. A circular disc of diameter 20 cm is divided equally into 5 sectors. Find the area and perimeter of one of the sectors. [Use $\pi = 3.14$] **[3]**

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★★★ END OF QUESTION PAPER · ALL THE BEST ★★★